

PHYS 357: Honours Quantum Physics 1

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August 31st, 2026

Abstract

Contents

1	Introduction	1
2	Mathematical Background	1
2.1	Taylor expansion	1
3	Stern-Gerlach Experiments	2
3.1	The classical picture	2
3.1.1	Angular momentum	2
3.1.2	Magnetic moment of a circulating charge	2
3.1.3	Gyromagnetic ratio	2
3.2	Spin	3
3.2.1	Intrinsic moment and the g-factor	3
3.2.2	Magnetons	3
3.2.3	Values for the electron, proton and neutron	4
3.3	The Stern-Gerlach apparatus	5
3.3.1	Force in an inhomogeneous field	5
3.3.2	Silver atoms	5
3.3.3	The classical prediction	6
3.3.4	The observed result	6
3.4	Sequential Stern-Gerlach experiments	7
3.4.1	Experiment 1: SGz then SGz	7
3.4.2	Experiment 2: SGz then SGx	8
3.4.3	Experiment 3: SGz, SGx, then SGz	8
3.4.4	Experiment 4: SGz, modified SGx, then SGz	8
4	Quantum States as Vectors	8
4.1	The vector space of spin states	8
4.1.1	Superposition	8
4.1.2	Bras and the inner product	9
4.1.3	Completeness	9
4.2	Extracting probabilities	9
4.2.1	Probability amplitudes	9
4.2.2	Normalization	10
4.2.3	Global phase	10
4.3	Fixing the basis states	11
4.3.1	The state $ +\hat{x}\rangle$	11
4.3.2	Experiment 5: measuring S_y	11
4.3.3	The state $ +\hat{y}\rangle$ and the phase convention	11
4.4	Statistics of measurements	12
4.4.1	Expectation value	12
4.4.2	Uncertainty	13
4.5	Generalization to n -level systems	14
4.5.1	Orthonormality and completeness	14
4.5.2	Observables and the Born rule	14

4.5.3	Expectation value as an operator	15
5	Rotation of Basis States and Matrix Mechanics	15
5.1	States as column vectors	15
5.2	Operators as matrices	16
5.3	The spin operators	17
5.4	The rotation operator	18
5.4.1	Rotations about $\hat{z}$	18
5.4.2	Rotation as a change of basis	19
6	Change of Basis in n-Level Systems	20
6.1	Two bases for the same space	20
6.2	The transformation matrix	20
6.3	Eigenstates and eigenvalues	21
6.4	Operators in another basis	21
A	Appendix	22
B	Solutions	22
C	Useful Links	24

1 Introduction

Textbook used is JS Townsend, *A Modern Approach to Quantum Mechanics*, 2nd ed., University Science Books 2012. Course grading is divided as follows:

- 10% problem sets (must be handwritten scans).
- 40% quizzes (on Wednesdays, best 8 of 12).
- 20% midterms (best of 2, October 2nd and November 6th).
- 30% final exam.

2 Mathematical Background

2.1 Taylor expansion

Definition 1 (Taylor series). *A function f that is smooth near a equals the power series*

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n, \quad (1)$$

where $f^{(n)}$ is the n th derivative. Expanding about $a = 0$ gives the Maclaurin series.

Result 1 (Series used in this course).

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}, \quad \cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}, \quad \sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}, \quad (2)$$

together with the binomial theorem $(1+u)^N = \sum_{n=0}^N \binom{N}{n} u^n$ for integer N .

Remark 1. Comparing the three series gives Euler's formula $e^{i\theta} = \cos \theta + i \sin \theta$, hence

$$\cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2}, \quad \sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}. \quad (3)$$

These two are used constantly when rotation matrices are written out.

Problem 1 (The exponential as a limit). Show that $\lim_{N \rightarrow \infty} (1 + x/N)^N = e^x$.

Remark 2. This limit is what lets a rotation be written as an exponential. A finite rotation is N infinitesimal ones stacked up,

$$R(\theta \hat{n}) = \lim_{N \rightarrow \infty} \left(\mathbb{1} - \frac{i}{\hbar} \frac{\theta}{N} \hat{n} \cdot \vec{S} \right)^N = e^{-i\theta \hat{n} \cdot \vec{S} / \hbar}. \quad (4)$$

August 31 -
September 9,
2026

3 Stern-Gerlach Experiments

3.1 The classical picture

3.1.1 Angular momentum

Definition 2 (Orbital angular momentum). *The orbital angular momentum of a particle is*

$$\vec{L} = \vec{r} \times \vec{p}, \quad (5)$$

where $\vec{r}$ is the position vector and $\vec{p} = m\vec{v}$ is the linear momentum.

For motion confined to the x - y plane, only the z component survives. On a circular path of radius r with speed v ,

$$L_z = mvr.$$

Definition 3 (Angular momentum of a rigid body). *For a rigid body¹ rotating about a fixed axis this is*

$$L = I_0\omega,$$

with I_0 the moment of inertia and ω the angular velocity.

Remark 3. *The moment of inertia I_0 measures how mass is distributed relative to the rotation axis.*

3.1.2 Magnetic moment of a circulating charge

Definition 4 (Magnetic moment). *A magnetic moment is a measure of the net circulation of electric charge. It is a vector quantity that determines the torque a magnet will experience in an external magnetic field.*

The magnetic moment of a point particle of mass m and charge q in a circular orbit of radius r and period T is

$$\mu = \frac{IA}{c} = \frac{1}{c} \left(\frac{qv}{2\pi r} \right) (\pi r^2) = \frac{q}{2mc} mvr = \frac{q}{2mc} L_z,$$

where I is the current and A is the area enclosed by the orbit. A single circulating charge counts as a steady current $I = q/T = qv/2\pi r$ because it passes any given point on the loop once per period.

3.1.3 Gyromagnetic ratio

Definition 5 (Gyromagnetic ratio). *The gyromagnetic ratio γ is the constant of proportionality between the magnetic moment and the angular momentum,*

$$\vec{\mu} = \gamma \vec{L}. \quad (6)$$

For the circulating charge above,

$$\gamma = \frac{q}{2mc}. \quad (7)$$

¹A rigid body is one that does not deform under the action of external forces.

²Here c is in Gaussian units.

Remark 4. *This is a purely classical result, since the charge is treated as a point particle on a definite orbit with a definite radius and speed, and nothing is quantized. It matters here because measuring $\vec{\mu}$ is how we measure $\vec{L}$, and it is what the Stern-Gerlach apparatus exploits.*

3.2 Spin

Definition 6 (Spin). *Spin $\vec{S}$ is an intrinsic angular momentum carried by a particle. It is a fixed property of the particle type, like charge or mass, and does not come from any motion through space. A stationary electron alone in empty space still has spin.*

In quantum mechanics the spin $\vec{S}$ replaces the orbital angular momentum $\vec{L}$ of the classical picture.

3.2.1 Intrinsic moment and the g-factor

The magnetic moment is still proportional to the angular momentum, but the gyromagnetic ratio picks up an extra dimensionless factor,

$$\vec{\mu} = \gamma \vec{S}, \quad \gamma = \frac{qg}{2mc}, \quad (8)$$

where g is the g-factor of the particle.

Definition 7 (g-factor). *The g-factor of a particle³ is the dimensionless number relating its magnetic moment to its spin angular momentum. It measures how much stronger or weaker the moment is than the classical value $\gamma = q/2mc$, which corresponds to $g = 1$.*

3.2.2 Magnetons

Definition 8 (Planck constant). *The Planck constant h is the fundamental constant of quantum mechanics. It sets the scale at which quantization matters, and relates the energy of a photon to its frequency by $E = h\nu$. Its value is*

$$h \approx 6.626 \times 10^{-34} \text{ J s.}$$

It has units of $J \cdot s$, which is the same as angular momentum.

Definition 9 (Reduced Planck constant).

$$\hbar = \frac{h}{2\pi}. \quad (9)$$

It carries units of angular momentum, so $[\hbar] = [L_z] = \frac{kg \cdot m^2}{s}$. Its value is $\hbar = 1.055 \times 10^{-27} \text{ erg s} = 6.582 \times 10^{-16} \text{ eV s}$.

³Its value must be measured or derived from relativistic quantum theory.

Writing $\vec{S} = \hbar(\vec{S}/\hbar)$ with $\vec{S}/\hbar$ dimensionless, the $\hbar$ joins the gyromagnetic ratio. The combination $q\hbar/2mc$ then carries the units and sets the natural size of the moment at a given mass scale.

Definition 10 (Bohr magneton). *The Bohr magneton is the natural unit of magnetic moment for an electron,*

$$\mu_B = \frac{|e|\hbar}{2m_e c}. \quad (10)$$

Definition 11 (Nuclear magneton). *The nuclear magneton is the same combination with the proton mass, and is the natural unit for nucleons,*

$$\mu_N = \frac{|e|\hbar}{2m_p c}. \quad (11)$$

Since $m_p/m_e \approx 1836$, we have $\mu_N \approx \mu_B/1836$.

Definition 12 (Nucleons). *The subatomic particles that make up the nucleus of an atom, specifically protons and neutrons.*

The electron moment is then written as

$$\vec{\mu}_e = g_e \mu_B \frac{\vec{S}}{\hbar}, \quad (12)$$

and similarly for the proton and neutron with μ_N in place of μ_B .

3.2.3 Values for the electron, proton and neutron

Particle	Charge q	Mass m	g -factor	Gyromagnetic ratio γ
Electron	$- e $	m_e	$g_e \approx 2$	$g_e \mu_B / \hbar$
Proton	$+ e $	m_p	$g_p \approx 5.59$	$g_p \mu_N / \hbar$
Neutron	0	m_p	$g_n \approx -3.83$	$g_n \mu_N / \hbar$

Table 1: Charges, masses, g -factors and gyromagnetic ratios.

Remark 5. *The classical derivation gives $\mu = (q/2mc)L_z$, which vanishes when $q = 0$. The neutron is neutral yet has a nonzero moment, so its $\vec{\mu}$ cannot come from a single circulating point charge. This is a first sign that the classical picture fails.*

Remark 6. *Since $m_p \gg m_e$, we get $\mu_N \ll \mu_B$ and therefore $|\vec{\mu}_p| \ll |\vec{\mu}_e|$. Stern-Gerlach deflection of nuclei is much weaker than for electrons.*

3.3 The Stern-Gerlach apparatus

Definition 13 (Stern-Gerlach apparatus). *A device that measures one component of a particle's magnetic moment. A beam of neutral atoms passes through a magnet shaped to produce a strong field gradient along a chosen axis, here z . The gradient deflects each atom by an amount proportional to μ_z , and a screen records where it lands. Since $\vec{\mu} = \gamma\vec{S}$, the position on the screen is a direct measurement of S_z .*

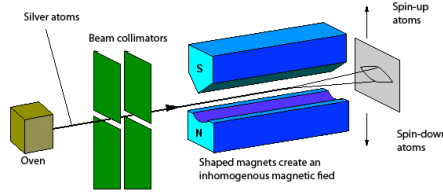


Figure 1: Illustration of the Stern-Gerlach apparatus.

3.3.1 Force in an inhomogeneous field

A magnetic moment in a field has potential energy

$$U = -\vec{\mu} \cdot \vec{B}, \quad (13)$$

so the force on it is

$$\vec{F} = -\vec{\nabla}U = \vec{\nabla}(\vec{\mu} \cdot \vec{B}), \quad F_z = -\frac{\partial U}{\partial z} = \mu_z \frac{\partial B_z}{\partial z}.$$

Definition 14 (Deflection). *The displacement of an atom on the screen, perpendicular to the beam axis, produced by F_z acting over the length of the magnet. It is proportional to μ_z , so the pattern on the screen is a record of the moments in the beam.*

Remark 7. *A uniform field gives no force, only a torque. Thus, the apparatus needs a field gradient to deflect anything.*

3.3.2 Silver atoms

Silver is used because its net orbital angular momentum vanishes, $\vec{L} = 0$, while $\vec{S} \neq 0$. The entire magnetic moment comes from a single unpaired valence electron, so the atom behaves as one electron spin. For the electron,

$$\mu_z = -\frac{g_e |e|}{2m_e c} S_z,$$

and therefore

$$F_z = \mu_z \frac{\partial B_z}{\partial z} = -g_e \frac{|e|}{2m_e c} S_z \frac{\partial B_z}{\partial z} = g_e \frac{|e|}{2m_e c} \left| \frac{\partial B_z}{\partial z} \right| S_z,$$

where the last equality assumes $\partial B_z / \partial z < 0$.

Result 2.

$$F_z \propto S_z.$$

The deflection of an atom measures the z component of its spin directly.

3.3.3 The classical prediction

Spin is a vector,

$$\vec{S} = (S_x, S_y, S_z)^T, \quad S_z = |\vec{S}| \cos \theta.$$

Remark 8. The spins leaving the oven point in random directions, so θ takes every value in $[0, \pi]$ across the beam and $S_z = |\vec{S}| \cos \theta$ takes every value in $[-|\vec{S}|, |\vec{S}|]$. Since $F_z \propto S_z$, each atom is deflected differently and the screen should show a continuous smear, but it shows two spots.

3.3.4 The observed result

Result 3 (Quantization of spin). The electron spin angular momentum is quantized:

$$S_z = \pm \frac{\hbar}{2}. \quad (14)$$

The beam splits into two spots.

Definition 15 (Spin quantum number). The spin quantum number s labels the magnitude of a particle's spin and is fixed by the particle type. It takes the values $s = 0, \frac{1}{2}, 1, \frac{3}{2}, \dots$, and a measurement of S_z then gives one of $2s + 1$ outcomes,

$$S_z = m\hbar, \quad m = -s, -s + 1, \dots, +s. \quad (15)$$

The electron has $s = \frac{1}{2}$, so $S_z = \pm \hbar/2$ and the beam splits in two. A spin-1 particle would split into three beams, $S_z = +\hbar, 0, -\hbar$.

Remark 9. Everything in these notes is the $s = \frac{1}{2}$ case, so the state space is two-dimensional. Note also that S_z is a component, while the magnitude is $|\vec{S}| = \hbar\sqrt{s(s+1)}$, which for the electron is $\hbar\sqrt{3}/2$, not $\hbar/2$.

Definition 16 (Basis state). One of a set of states $\{|j\rangle\}$ that are mutually orthonormal and span the space, so that every state is a unique linear combination of them. For spin- $\frac{1}{2}$ the two outcomes of any one measurement, $\{|\pm\hat{n}\rangle\}$, form a basis.

Definition 17 (Ket notation). A ket $|\cdot\rangle$ labels a quantum state as a vector in a vector space. For electron spin the two measurement outcomes give the basis states

$$|\uparrow\rangle \text{ for } S_z = +\frac{\hbar}{2}, \quad |\downarrow\rangle \text{ for } S_z = -\frac{\hbar}{2}.$$

The space is two-dimensional, so any spin state is a linear combination of the two.

Remark 10. From here on we follow Townsend and write $|+\hat{z}\rangle \equiv |\uparrow\rangle$ and $|-\hat{z}\rangle \equiv |\downarrow\rangle$. The same labels work for any axis: $|\pm\hat{x}\rangle$ are the states with $S_x = \pm\hbar/2$.

3.4 Sequential Stern-Gerlach experiments

Proposition 1. We write SG_z for a device whose gradient points along z , and SG_x for the same device rotated by 90° about the beam axis.

Definition 18 (Filter). Each device sends atoms into two beams. A filter is a device that blocks one of the beams.

Remark 11. A beam straight from the oven always splits 50/50 in any SG device, whatever its orientation.

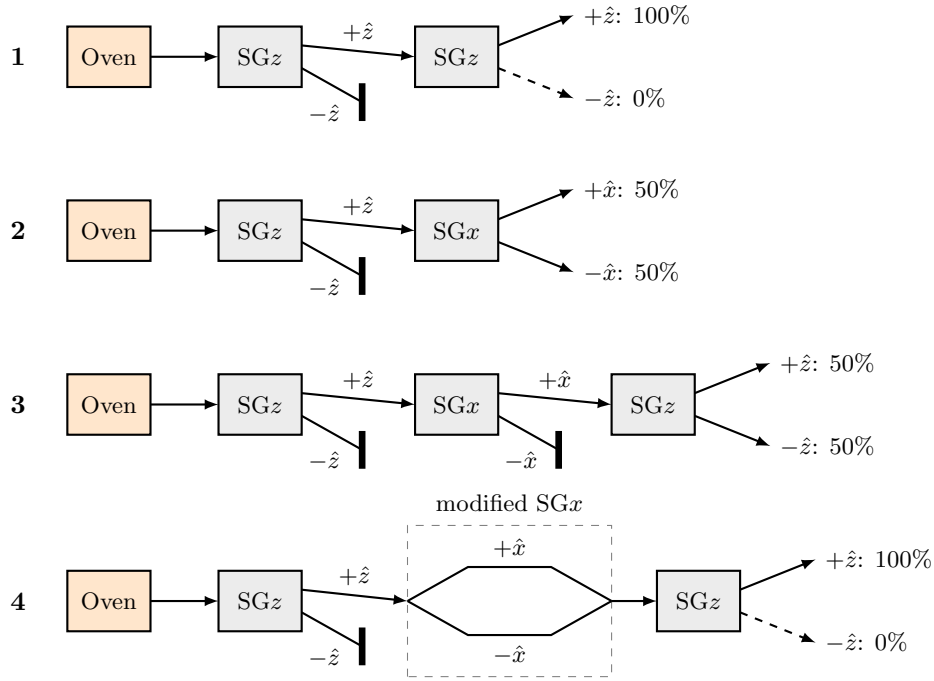


Figure 2: Sequential Stern-Gerlach experiments. Thick bars block a beam. Percentages are the fractions of atoms entering the last device. Every oven starts with N_0 atoms.

3.4.1 Experiment 1: SG_z then SG_z

Filter the $+\hat{z}$ beam and send it into a second SG_z . Every atom comes out $+\hat{z}$, none come out $-\hat{z}$. The measurement is repeatable: once an atom is found with $S_z = +\hbar/2$, measuring S_z again gives the same answer. The atom leaves the first device in the state $|+\hat{z}\rangle$. So we end with $N_0/2$ atoms in $|+\hat{z}\rangle$.

3.4.2 Experiment 2: SG_z then SG_x

Filter the $+\hat{z}$ beam and send it into an SG_x. Half the atoms come out $+\hat{x}$ and half come out $-\hat{x}$. Knowing $S_z = +\hbar/2$ tells us nothing about S_x . We end up with $N_0/4$ in $|+\hat{x}\rangle$ and $N_0/4$ in $|-\hat{x}\rangle$.

3.4.3 Experiment 3: SG_z, SG_x, then SG_z

Filter $+\hat{z}$, then filter $+\hat{x}$, then measure S_z again. The beam splits 50/50 into $+\hat{z}$ and $-\hat{z}$ at the end, even though every atom was filtered to $+\hat{z}$ at the start. The S_x measurement wiped out the earlier S_z information. Blocking $+\hat{x}$ instead of $-\hat{x}$ gives the same 50/50 result. We end with $N_0/8$ in $|+\hat{z}\rangle$ and $N_0/8$ in $|-\hat{z}\rangle$.

Result 4. *An atom with a definite value of S_x does not have a definite value of S_z . Measuring S_x destroys what we knew about S_z , so no state has definite values of both at once.*

3.4.4 Experiment 4: SG_z, modified SG_x, then SG_z

Definition 19 (Modified SG device). *An SG device that splits the beam according to S_n and then recombines the two beams, so atoms with $S_n = \pm\hbar/2$ emerge at the same position even though they took two different paths.*

Filter $+\hat{z}$, send it through a modified SG_x with neither path blocked, then measure S_z . Every atom comes out $+\hat{z}$: 100%, not the 50/50 of Experiment 3. All $N_0/2$ atoms end in $|+\hat{z}\rangle$.

Result 5. *The modified SG_x cannot be sending $N_0/4$ through as $|+\hat{x}\rangle$ and $N_0/4$ as $|-\hat{x}\rangle$: Experiment 3 showed each of those alone gives 50/50 at the end. Instead the two paths interfere, as for water or light waves, cancelling the possibility of $-\hat{z}$. No path can be assigned to an atom.*

4 Quantum States as Vectors

4.1 The vector space of spin states

4.1.1 Superposition

Definition 20 (Superposition). *A general spin state is a linear combination of the basis kets,*

$$|\psi\rangle = c_+ |+\hat{z}\rangle + c_- |-\hat{z}\rangle, \quad c_{\pm} \in \mathbb{C}. \quad (16)$$

The pair $\{|+\hat{z}\rangle, |-\hat{z}\rangle\}$ is a basis, and the coefficients $c_{\pm}$ are complex numbers.

September 11-
18, 2026

4.1.2 Bras and the inner product

Definition 21 (Bra). *Every ket $|\psi\rangle$ has a matching bra $\langle\psi|$. Putting a bra in front of a ket gives a complex number, the inner product*

$$\langle\phi| |\psi\rangle \equiv \langle\phi|\psi\rangle \in \mathbb{C}.$$

Definition 22 (Orthonormal basis). *The basis kets are orthonormal:*

$$\langle+\hat{z}| +\hat{z}\rangle = \langle-\hat{z}| -\hat{z}\rangle = 1, \quad \langle+\hat{z}| -\hat{z}\rangle = \langle-\hat{z}| +\hat{z}\rangle = 0. \quad (17)$$

4.1.3 Completeness

Definition 23 (Projector). *The operator $P_j = |j\rangle\langle j|$ built from a basis state. Acting on $|\psi\rangle = \sum_k c_k |k\rangle$ it keeps only the $|j\rangle$ part,*

$$P_j |\psi\rangle = c_j |j\rangle,$$

since $\langle j|k\rangle = \delta_{jk}$.

Proposition 2 (Completeness relation). *Since any state can be expanded in the basis $\{|+\hat{z}\rangle, |-\hat{z}\rangle\}$, the sum of the projectors onto the basis states is the identity:*

$$|+\hat{z}\rangle\langle+\hat{z}| + |-\hat{z}\rangle\langle-\hat{z}| = \mathbf{1}. \quad (18)$$

Inserting the identity in front of any state gives its expansion directly:

$$|\psi\rangle = (|+\hat{z}\rangle\langle+\hat{z}| + |-\hat{z}\rangle\langle-\hat{z}|) |\psi\rangle = \langle+\hat{z}|\psi\rangle |+\hat{z}\rangle + \langle-\hat{z}|\psi\rangle |-\hat{z}\rangle,$$

so the coefficients $c_{\pm}$ of the superposition are the inner products $\langle\pm\hat{z}|\psi\rangle$.

4.2 Extracting probabilities

4.2.1 Probability amplitudes

Proposition 3. *To pull out c_+ , act on $|\psi\rangle$ with $\langle+\hat{z}|$:*

$$\langle+\hat{z}|\psi\rangle = c_+ \underbrace{\langle+\hat{z}|+\hat{z}\rangle}_{=1} + c_- \underbrace{\langle+\hat{z}|-\hat{z}\rangle}_{=0} = c_+.$$

The same steps with $\langle-\hat{z}|$ give c_- . So

$$c_{\pm} = \langle\pm\hat{z}|\psi\rangle. \quad (19)$$

Definition 24 (Probability amplitude). *The coefficient $c_{\pm} = \langle\pm\hat{z}|\psi\rangle$ is the probability amplitude for measuring $S_z = \pm\hbar/2$. The probability itself is the modulus squared,*

$$P(S_z = \pm\frac{\hbar}{2}) = |c_{\pm}|^2 = |\langle\pm\hat{z}|\psi\rangle|^2. \quad (20)$$

Principle 1 (Born rule). *The probability that a system in the state $|\psi\rangle$ is found in the state $|\phi\rangle$ is*

$$P(\phi|\psi) = |\langle\phi|\psi\rangle|^2 \in [0, 1]. \quad (21)$$

Remark 12. *Why is the Born rule true? There is no deeper derivation: it is a postulate that agrees with every experiment. As David Mermin put it, "shut up and calculate."*

4.2.2 Normalization

Proposition 4. *A system in $|\psi\rangle$ is found in $|\psi\rangle$ with certainty, so*

$$P(\psi|\psi) = |\langle\psi|\psi\rangle|^2 = 1.$$

To find what a bra equals to, write it with unknown coefficients, $\langle\psi| = c'_+ \langle+\hat{z}| + c'_- \langle-\hat{z}|$. Then

$$\langle\psi|\psi\rangle = (c'_+ \langle+\hat{z}| + c'_- \langle-\hat{z}|) (c_+ |+\hat{z}\rangle + c_- |-\hat{z}\rangle) = c'_+ c_+ + c'_- c_-,$$

since the cross terms vanish by orthonormality. The two outcomes $S_z = \pm\hbar/2$ must have probabilities $|c_+|^2$ and $|c_-|^2$ that add to 1. For $\langle\psi|\psi\rangle$ to be that sum, each product $c'_\pm c_\pm$ must be real and equal to $|c_\pm|^2$. This forces $c'_\pm = c_\pm^*$.

Result 6. *Going from a ket to its bra, conjugate the coefficients:*

$$\langle\psi| = c_+^* \langle+\hat{z}| + c_-^* \langle-\hat{z}|. \quad (22)$$

In particular $\langle\psi|\pm\hat{z}\rangle = c_\pm^ = \langle\pm\hat{z}|\psi\rangle^*$. More generally, $\langle\phi|\psi\rangle = \langle\psi|\phi\rangle^*$.*

Definition 25 (Normalization condition).

$$\langle\psi|\psi\rangle = \langle\psi|+\hat{z}\rangle \langle+\hat{z}|\psi\rangle + \langle\psi|-\hat{z}\rangle \langle-\hat{z}|\psi\rangle = |c_+|^2 + |c_-|^2 = 1. \quad (23)$$

In words, $P(S_z = +\hbar/2) + P(S_z = -\hbar/2) = 1$: the atom is always found in one of the two beams.

Example. *Take*

$$|\psi\rangle = \frac{1}{\sqrt{3}} |+\hat{z}\rangle + i\sqrt{\frac{2}{3}} |-\hat{z}\rangle.$$

The amplitude for $S_z = +\hbar/2$ is

$$\langle+\hat{z}|\psi\rangle = \frac{1}{\sqrt{3}} \underbrace{\langle+\hat{z}|+\hat{z}\rangle}_{=1} + i\sqrt{\frac{2}{3}} \underbrace{\langle+\hat{z}|-\hat{z}\rangle}_{=0} = \frac{1}{\sqrt{3}},$$

so $P(S_z = +\hbar/2) = 1/3$. In the same way $\langle-\hat{z}|\psi\rangle = i\sqrt{2/3}$, so $P(S_z = -\hbar/2) = |i|^2 \cdot 2/3 = 2/3$. The two add to 1, so the state is normalized.

4.2.3 Global phase

Proposition 5. *Take two kets that differ only by an overall phase, $|\psi\rangle = e^{i\alpha} |\psi'\rangle$ with $\alpha \in \mathbb{R}$. For any state $|\phi\rangle$,*

$$P(\phi|\psi) = |\langle\phi|\psi\rangle|^2 = |e^{i\alpha} \langle\phi|\psi'\rangle|^2 = |\langle\phi|\psi'\rangle|^2 = P(\phi|\psi'),$$

since $|e^{i\alpha}| = 1$.

Result 7. *$|\psi\rangle$ and $e^{i\alpha} |\psi\rangle$ give the same probabilities for every possible measurement, so they describe the same physical state. An overall (global) phase has no physical meaning and can be dropped.*

Remark 13. *A relative phase is different: it sits between the two terms of a superposition, so it cannot be pulled out front and dropped. It has physical consequences, as the next subsection shows.*

4.3 Fixing the basis states

4.3.1 The state $|+\hat{x}\rangle$

In Experiment 3, atoms in $|+\hat{x}\rangle$ enter an SG z and split 50/50:

$$P(S_z = \pm \frac{\hbar}{2}) = |\langle \pm \hat{z} | +\hat{x} \rangle|^2 = \frac{1}{2}.$$

So both amplitudes have modulus $1/\sqrt{2}$, and each can carry its own phase:

$$|+\hat{x}\rangle = \frac{1}{\sqrt{2}} e^{i\delta_+} |+\hat{z}\rangle + \frac{1}{\sqrt{2}} e^{i\delta_-} |-\hat{z}\rangle = \frac{e^{i\delta_+}}{\sqrt{2}} (|+\hat{z}\rangle + e^{i\delta} |-\hat{z}\rangle), \quad \delta = \delta_- - \delta_+,$$

with $\delta_{\pm} \in \mathbb{R}$. Dropping the global phase $e^{i\delta_+}$ leaves

$$|+\hat{x}\rangle = \frac{1}{\sqrt{2}} (|+\hat{z}\rangle + e^{i\delta} |-\hat{z}\rangle).$$

Remark 14. *The relative phase δ survives, and Experiment 3 alone does not fix it. It is pinned down by combining more measurements (for example along y) with a phase convention.*

4.3.2 Experiment 5: measuring S_y

Definition 26 (Cyclic permutation). *A cyclic permutation of the axes relabels them as $x \rightarrow y$, $y \rightarrow z$, $z \rightarrow x$ (or the reverse). It is a rotation of the coordinate system, so no physics changes under it.*

Let Experiment 5 be a cyclic permutation of Experiment 3, SG x becomes SG z and SG z becomes SG y . Now the process is as follows: filter $+\hat{z}$, then measure S_y .

Proposition 6. *By symmetry it must split 50/50 as well, so every cross-axis overlap has the same modulus:*

$$|\langle +\hat{x} | +\hat{z} \rangle|^2 = |\langle +\hat{y} | +\hat{z} \rangle|^2 = |\langle -\hat{x} | +\hat{y} \rangle|^2 = \frac{1}{2}.$$

The same argument as for $|+\hat{x}\rangle$ then gives

$$|+\hat{y}\rangle = \frac{e^{i\gamma_+}}{\sqrt{2}} (|+\hat{z}\rangle + e^{i\gamma} |-\hat{z}\rangle), \quad \gamma = \gamma_- - \gamma_+.$$

4.3.3 The state $|+\hat{y}\rangle$ and the phase convention

Proposition 7. *Dropping the global phases, $\gamma_+ = \delta_+ = 0$, we obtain equations for x and y in terms of z :*

$$|\pm\hat{x}\rangle = \frac{1}{\sqrt{2}} (|+\hat{z}\rangle \pm e^{i\delta} |-\hat{z}\rangle), \quad |\pm\hat{y}\rangle = \frac{1}{\sqrt{2}} (|+\hat{z}\rangle \pm e^{i\gamma} |-\hat{z}\rangle).$$

So using orthonormality of the $\hat{z}$ basis,

$$\langle +\hat{x} | +\hat{y} \rangle = \frac{1}{2} (1 + e^{i(\gamma-\delta)}), \quad |\langle +\hat{x} | +\hat{y} \rangle|^2 = \frac{1}{2} (1 + \cos(\gamma - \delta)).$$

Requiring this to equal 1/2 forces $\cos(\gamma - \delta) = 0$, that is $\gamma - \delta = \pm\pi/2$. The remaining freedom is convention.

Result 8 (Phase convention). *Take $\delta = 0$ and $\gamma = +\pi/2$. Then*

$$|\pm\hat{x}\rangle = \frac{1}{\sqrt{2}} (|+\hat{z}\rangle \pm |-\hat{z}\rangle), \quad |\pm\hat{y}\rangle = \frac{1}{\sqrt{2}} (|+\hat{z}\rangle \pm i |-\hat{z}\rangle). \quad (24)$$

The bras are then

$$\langle \pm\hat{x} | = \frac{1}{\sqrt{2}} (\langle +\hat{z} | \pm \langle -\hat{z} |), \quad \langle \pm\hat{y} | = \frac{1}{\sqrt{2}} (\langle +\hat{z} | \mp i \langle -\hat{z} |). \quad (25)$$

Remark 15. *Complex coefficients are required, not just allowed. With real coefficients the relative phases could only be 0 or π , so $\gamma - \delta = \pm\pi/2$ would be impossible.*

Example. For $|\psi\rangle = \sqrt{\frac{1}{3}} |+\hat{z}\rangle + i\sqrt{\frac{2}{3}} |-\hat{z}\rangle$, compute $\langle S_y \rangle$.
The amplitudes are

$$\begin{aligned} \langle +\hat{y} | \psi \rangle &= \sqrt{\frac{1}{3}} \langle +\hat{y} | +\hat{z} \rangle + i\sqrt{\frac{2}{3}} \langle +\hat{y} | -\hat{z} \rangle = \frac{1}{\sqrt{2}} \left(\sqrt{\frac{1}{3}} + i\sqrt{\frac{2}{3}} (-i) \right) = \frac{1}{\sqrt{2}} \left(\sqrt{\frac{1}{3}} + \sqrt{\frac{2}{3}} \right), \\ \langle -\hat{y} | \psi \rangle &= \sqrt{\frac{1}{3}} \langle -\hat{y} | +\hat{z} \rangle + i\sqrt{\frac{2}{3}} \langle -\hat{y} | -\hat{z} \rangle = \frac{1}{\sqrt{2}} \left(\sqrt{\frac{1}{3}} + i\sqrt{\frac{2}{3}} (i) \right) = \frac{1}{\sqrt{2}} \left(\sqrt{\frac{1}{3}} - \sqrt{\frac{2}{3}} \right). \end{aligned}$$

Computing the probabilities

$$P(S_y = +\frac{\hbar}{2}) = |\langle +\hat{y} | \psi \rangle|^2 = \frac{1}{2} + \frac{\sqrt{2}}{3}, \quad P(S_y = -\frac{\hbar}{2}) = |\langle -\hat{y} | \psi \rangle|^2 = \frac{1}{2} - \frac{\sqrt{2}}{3},$$

which add to 1 as they must. Then

$$\langle S_y \rangle = \frac{\hbar}{2} \left[\left(\frac{1}{2} + \frac{\sqrt{2}}{3} \right) - \left(\frac{1}{2} - \frac{\sqrt{2}}{3} \right) \right] = \frac{\sqrt{2}}{3} \hbar.$$

The factor i that did nothing for S_z matters here: it is a relative phase, and it is exactly what tilts the spin toward $+\hat{y}$.

4.4 Statistics of measurements

4.4.1 Expectation value

Definition 27 (Expectation value). *The expectation value of S_z in the state $|\psi\rangle$ is the average outcome weighted by the probabilities,*

$$\langle S_z \rangle = \left(\frac{\hbar}{2} \right) |\langle +\hat{z} | \psi \rangle|^2 + \left(-\frac{\hbar}{2} \right) |\langle -\hat{z} | \psi \rangle|^2. \quad (26)$$

Remark 16. For N measurements on identically prepared atoms, with N_+ giving $+\hbar/2$ and N_- giving $-\hbar/2$, the measured average is

$$\langle S_z \rangle_N = \frac{\hbar}{2} \frac{N_+ - N_-}{N},$$

which tends to $\langle S_z \rangle$ as $N \rightarrow \infty$.

Remark 17 (Shot noise). For a finite number N of measurements, the sample average deviates from the expectation value by

$$|\langle S_z \rangle_N - \langle S_z \rangle| \sim \frac{\Delta S_z}{\sqrt{N}}.$$

This spread is intrinsic, not a flaw of the apparatus: it appears whenever a state with a definite value of one observable (e.g. S_x for $|\hat{x}\rangle$) is measured for an incompatible observable (e.g. S_z).

4.4.2 Uncertainty

Definition 28 (Uncertainty). The uncertainty ΔS_z is the standard deviation of the measurement outcomes,

$$(\Delta S_z)^2 = \langle (S_z - \langle S_z \rangle)^2 \rangle = \langle S_z^2 \rangle - \langle S_z \rangle^2, \quad (27)$$

where

$$\langle S_z^2 \rangle = \left(\frac{\hbar}{2}\right)^2 |\langle +\hat{z}|\psi \rangle|^2 + \left(-\frac{\hbar}{2}\right)^2 |\langle -\hat{z}|\psi \rangle|^2. \quad (28)$$

Example. For $|\hat{x}\rangle$, both probabilities are $1/2$, so

$$\langle S_z \rangle = \frac{\hbar}{2} \cdot \frac{1}{2} - \frac{\hbar}{2} \cdot \frac{1}{2} = 0, \quad \langle S_z^2 \rangle = \left(\frac{\hbar}{2}\right)^2 \cdot \frac{1}{2} + \left(-\frac{\hbar}{2}\right)^2 \cdot \frac{1}{2} = \frac{\hbar^2}{4},$$

and therefore $\Delta S_z = \hbar/2$. The average is zero but every single measurement gives $\pm\hbar/2$, so the spread is maximal.

Example. For $|\psi\rangle = \sqrt{\frac{1}{3}}|\hat{z}\rangle + i\sqrt{\frac{2}{3}}|-\hat{z}\rangle$, the probabilities are $1/3$ and $2/3$, so

$$\langle S_z \rangle = \frac{\hbar}{2} \cdot \frac{1}{3} - \frac{\hbar}{2} \cdot \frac{2}{3} = -\frac{\hbar}{6}, \quad \langle S_z^2 \rangle = \frac{\hbar^2}{4} \left(\frac{1}{3} + \frac{2}{3}\right) = \frac{\hbar^2}{4}.$$

Then

$$(\Delta S_z)^2 = \frac{\hbar^2}{4} - \frac{\hbar^2}{36} = \frac{9\hbar^2 - \hbar^2}{36} = \frac{2}{9}\hbar^2, \quad \Delta S_z = \frac{\sqrt{2}}{3}\hbar.$$

Remark 18. For spin- $\frac{1}{2}$, $\langle S_z^2 \rangle$ is always equal to $\frac{\hbar^2}{4}$, since both outcomes square to the same number and the probabilities sum to 1.

4.5 Generalization to n -level systems

4.5.1 Orthonormality and completeness

A spin- $\frac{1}{2}$ particle is a two-level system, and each of $\{|\pm\hat{x}\rangle\}$, $\{|\pm\hat{y}\rangle\}$, $\{|\pm\hat{z}\rangle\}$ is a basis. An n -level system has a basis $\{|j\rangle\}$, $j = 1, \dots, n$.

Remark 19. The index j labels the outcomes of one chosen observable, not the coordinate axes. Picking S_x , S_y or S_z picks the basis; within that basis $j = 1, 2$ labels its two states, for example $|a_1\rangle = |+\hat{x}\rangle$ and $|a_2\rangle = |-\hat{x}\rangle$. A spin-1 particle has $n = 3$ outcomes, and an n -level atom has $j = 1, \dots, n$.

Definition 29 (Orthonormal and complete basis). The basis is orthonormal if

$$\langle i | j \rangle = \delta_{ij},$$

and complete if every state can be expanded in it,

$$|\psi\rangle = \sum_j c_j |j\rangle, \quad c_j = \langle j | \psi \rangle, \quad \langle \psi | = \sum_j c_j^* \langle j|. \quad (29)$$

Completeness is equivalent to

$$\mathbb{1} = \sum_{j=1}^n |j\rangle \langle j|. \quad (30)$$

Example. For $n = 2$, $\mathbb{1} = |+\hat{n}\rangle \langle +\hat{n}| + |-\hat{n}\rangle \langle -\hat{n}|$ for any $\hat{n} \in \{\hat{x}, \hat{y}, \hat{z}\}$, so

$$|\psi\rangle = \langle +\hat{x} | \psi \rangle |+\hat{x}\rangle + \langle -\hat{x} | \psi \rangle |-\hat{x}\rangle = \langle +\hat{y} | \psi \rangle |+\hat{y}\rangle + \langle -\hat{y} | \psi \rangle |-\hat{y}\rangle.$$

4.5.2 Observables and the Born rule

Definition 30 (Observable). An observable A has outcomes a_j with states $|a_j\rangle$ forming a complete orthonormal set. For spin- $\frac{1}{2}$, A could be S_x , S_y or S_z , with outcomes $\pm\hbar/2$. By the Born rule,

$$P(A = a_j) = |\langle a_j | \psi \rangle|^2. \quad (31)$$

Proposition 8. Normalization $\langle \psi | \psi \rangle = 1$ is the statement that the probabilities sum to 1.

Proof.

$$\langle \psi | \psi \rangle = \left(\sum_i c_i^* \langle a_i | \right) \left(\sum_j c_j |a_j\rangle \right) = \sum_{i,j} c_i^* c_j \delta_{ij} = \sum_j |c_j|^2 = \sum_j P(A = a_j) = 1.$$

Equivalently, inserting the identity, $\sum_j \langle \psi | a_j \rangle \langle a_j | \psi \rangle = \langle \psi | \left(\sum_j |a_j\rangle \langle a_j| \right) | \psi \rangle = \langle \psi | \psi \rangle$. $\square$

4.5.3 Expectation value as an operator

Proposition 9.

$$\langle A \rangle = \sum_j a_j |\langle a_j | \psi \rangle|^2 = \sum_j \langle \psi | a_j \rangle a_j \langle a_j | \psi \rangle = \langle \psi | \hat{A} | \psi \rangle, \quad \hat{A} = \sum_j a_j |a_j\rangle \langle a_j|. \quad (32)$$

Definition 31 (Uncertainty of an observable).

$$(\Delta A)^2 = \langle (A - \langle A \rangle)^2 \rangle = \langle A^2 \rangle - \langle A \rangle^2, \quad \langle A^2 \rangle = \langle \psi | \hat{A}^2 | \psi \rangle = \sum_j a_j^2 |c_j|^2. \quad (33)$$

5 Rotation of Basis States and Matrix Mechanics

September 21-,
2026

5.1 States as column vectors

Definition 32 (Representation of a state). *In the S_z basis, a ket is represented by the column of its components and a bra by the conjugate row, the adjoint:*

$$|\psi\rangle \xrightarrow{S_z} \psi = \begin{pmatrix} c_+ \\ c_- \end{pmatrix}, \quad \langle \psi | \xrightarrow{S_z} \psi^\dagger = (c_+^* \quad c_-^*), \quad (34)$$

with $c_\pm = \langle \pm \hat{z} | \psi \rangle$. Then

$$\langle \psi | \psi \rangle = \psi^\dagger \psi = |c_+|^2 + |c_-|^2. \quad (35)$$

Remark 20. Write $|\psi\rangle \xrightarrow{S_z} \psi$, not $|\psi\rangle = \psi$. The ket is basis-independent; the column depends on the basis.

Example.

$$|\pm \hat{x}\rangle \xrightarrow{S_z} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ \pm 1 \end{pmatrix}, \quad |\pm \hat{y}\rangle \xrightarrow{S_z} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ \pm i \end{pmatrix}.$$

Problem 2 (Other bases). *In the S_x basis, $|\psi\rangle \xrightarrow{S_x} \begin{pmatrix} \langle +\hat{x} | \psi \rangle \\ \langle -\hat{x} | \psi \rangle \end{pmatrix}$, and similarly for S_y . Write the column representing $|+\hat{z}\rangle$ in the S_z , S_x and S_y bases.*

Proposition 10 (Inner product as a matrix product). *If $|\phi\rangle \xrightarrow{S_z} \phi$ and $|\psi\rangle \xrightarrow{S_z} \psi$, then the inner product is a row times a column,*

$$\langle \phi | \psi \rangle = \phi^\dagger \psi, \quad (36)$$

and the Born rule reads $P(\phi | \psi) = |\phi^\dagger \psi|^2$.

Remark 21. If the state is given in another basis, first replace each basis ket by its S_z column and add. For $|\psi\rangle = a|+\hat{y}\rangle + b|-\hat{y}\rangle$,

$$\psi \xrightarrow{S_z} \frac{a}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix} + \frac{b}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} a+b \\ i(a-b) \end{pmatrix},$$

and $\psi^\dagger \psi = 1$ is worth checking before going further.

Problem 3 (A probability by matrix mechanics). For $|\psi\rangle = \sqrt{\frac{1}{5}}|+\hat{z}\rangle + i\sqrt{\frac{4}{5}}|-\hat{z}\rangle$, find $P(S_x = +\hbar/2)$ using matrix mechanics.

5.2 Operators as matrices

September 21,
2026

Definition 33 (Projection operator). $P_{\pm\hat{n}} = |\pm\hat{n}\rangle\langle\pm\hat{n}|$, so that $\mathbb{1} = P_{+\hat{n}} + P_{-\hat{n}}$ and $P_{+\hat{z}}|\psi\rangle = c_+|+\hat{z}\rangle$.

Definition 34 (Matrix of an operator). In a basis $\{|j\rangle\}$, an operator A is represented by the matrix

$$A_{ij} = \langle i|A|j\rangle. \quad (37)$$

Proposition 11 (Operator acting on a state). Acting with an operator on a ket becomes a matrix times a column vector,

$$A|\psi\rangle \rightarrow A\psi.$$

Proof. The k th entry of the column representing $A|\psi\rangle$ is its overlap with the k th basis state, $\langle k|A|\psi\rangle$. Insert $\mathbb{1} = \sum_j |j\rangle\langle j|$ between A and $|\psi\rangle$:

$$\langle k|A|\psi\rangle = \sum_j \underbrace{\langle k|A|j\rangle}_{A_{kj}} \underbrace{\langle j|\psi\rangle}_{c_j} = \sum_j A_{kj} c_j.$$

This is exactly row k of the matrix A multiplied by the column $\psi = (c_1, c_2, \dots)^T$, which is the rule for matrix-times-vector multiplication. $\square$

Example.

$$P_{+\hat{z}} \xrightarrow{S_z} \begin{pmatrix} \langle +\hat{z}|P_{+\hat{z}}|+\hat{z}\rangle & \langle +\hat{z}|P_{+\hat{z}}|-\hat{z}\rangle \\ \langle -\hat{z}|P_{+\hat{z}}|+\hat{z}\rangle & \langle -\hat{z}|P_{+\hat{z}}|-\hat{z}\rangle \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} c_+ \\ c_- \end{pmatrix} = \begin{pmatrix} c_+ \\ 0 \end{pmatrix}.$$

Proposition 12 (Properties of projectors). A projector satisfies $P_{\hat{n}}^2 = P_{\hat{n}}$, since $(|\hat{n}\rangle\langle\hat{n}|)(|\hat{n}\rangle\langle\hat{n}|) = |\hat{n}\rangle\langle\hat{n}|\hat{n}\rangle\langle\hat{n}| = |\hat{n}\rangle\langle\hat{n}|$. In the S_z basis,

$$P_{-\hat{z}} \xrightarrow{S_z} \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \quad \mathbb{1} = P_{+\hat{z}} + P_{-\hat{z}} \xrightarrow{S_z} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Definition 35 (Adjoint). Writing $A = \mathbb{1}A\mathbb{1} = \sum_{i,j} A_{ij}|i\rangle\langle j|$, the adjoint is

$$A^\dagger = \sum_{i,j} A_{ij}^* |j\rangle\langle i|, \quad (A^\dagger)_{ij} = A_{ji}^*. \quad (38)$$

Proposition 13 (Bra of a transformed state). *If $|\psi'\rangle = A|\psi\rangle$, then $\langle\psi'| = \langle\psi|A^\dagger$.*

Proposition 14 (Adjoint of a product).

$$(AB)^\dagger = B^\dagger A^\dagger. \quad (39)$$

Remark 22. *The order reverses, and $A^\dagger B^\dagger$ is a different matrix in general. For real entries the adjoint is just the transpose.*

5.3 The spin operators

Each spin component is built from the projectors onto its own basis, as the operator form of the expectation value requires:

$$\hat{S}_n = \frac{\hbar}{2} (|+\hat{n}\rangle\langle+\hat{n}| - |-\hat{n}\rangle\langle-\hat{n}|), \quad \hat{n} \in \{\hat{x}, \hat{y}, \hat{z}\}. \quad (40)$$

Its matrix in the S_z basis is $(\hat{S}_n)_{ij} = \langle i|\hat{S}_n|j\rangle$ with $i, j \in \{+\hat{z}, -\hat{z}\}$. Each entry is a difference of two products, one overlap from the bra side and one from the ket side, and the bra side is conjugated.

Problem 4 (The matrix of $\hat{S}_z$). *Find the matrix representation of $\hat{S}_z$ in the S_z basis.*

Problem 5 (The matrix of $\hat{S}_x$). *Find the matrix representation of $\hat{S}_x$ in the S_z basis.*

Problem 6 (The matrix of $\hat{S}_y$). *Find the matrix representation of $\hat{S}_y$ in the S_z basis.*

Result 9 (Pauli matrices).

$$\hat{S}_j = \frac{\hbar}{2}\sigma_j, \quad \sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad (41)$$

all in the S_z basis. Each is its own adjoint, and each squares to $\mathbb{1}$, so $\hat{S}_j^2 = \frac{\hbar^2}{4}\mathbb{1}$ for every j .

Remark 23 (Shortcut). *Instead of four separate brackets, write each projector as an outer product of a column with its conjugate row, for example*

$$|+\hat{x}\rangle\langle+\hat{x}| \xrightarrow{S_z} \frac{1}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad |-\hat{x}\rangle\langle-\hat{x}| \xrightarrow{S_z} \frac{1}{2} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix},$$

then subtract and multiply by $\hbar/2$.

Problem 7 (Expectation value and uncertainty by matrix mechanics). *For $|\psi\rangle = \sqrt{\frac{1}{5}}|+\hat{y}\rangle + i\sqrt{\frac{4}{5}}|-\hat{y}\rangle$, find $\langle S_z \rangle$ and ΔS_z using matrix mechanics.*

5.4 The rotation operator

Definition 36 (Rotation operator). $R(\theta\hat{n})$ rotates a state by θ about $\hat{n}$: $|\psi'\rangle = R(\theta\hat{n})|\psi\rangle$. For example, $R(\frac{\pi}{2}\hat{y})$ takes $|\pm\hat{z}\rangle$ to $|\pm\hat{x}\rangle$ (up to phase), so

$$R\left(\frac{\pi}{2}\hat{y}\right)(c_+|+\hat{z}\rangle + c_-|-\hat{z}\rangle) = c_+|+\hat{x}\rangle + c_-|-\hat{x}\rangle.$$

Remark 24. $R(\theta\hat{n})$ is a linear operator: $R(c_+|+\hat{z}\rangle + c_-|-\hat{z}\rangle) = c_+R|+\hat{z}\rangle + c_-R|-\hat{z}\rangle$, which is what the equation above uses.

Remark 25. The bra of $|\psi'\rangle$ is $\langle\psi|R^\dagger$, not $\langle\psi|R$.

Theorem 1. Rotation operators are unitary:

$$R^\dagger R = RR^\dagger = \mathbb{1}. \quad (42)$$

Proof. A rotation must preserve normalization: $\langle\psi'|\psi'\rangle = \langle\psi|R^\dagger R|\psi\rangle = 1$ for every $|\psi\rangle$ and every $R(\theta\hat{n})$, which forces $R^\dagger R = \mathbb{1}$. $\square$

Definition 37 (Unitary Operator). Unitary operators map physical (normalized) states to physical states. If a unitary U satisfies $U|\psi\rangle = c|\psi\rangle$, then $|c| = 1$, so $c = e^{i\varphi}$ is just a global phase and $U|\psi\rangle$ describes the same physical state as $|\psi\rangle$.

5.4.1 Rotations about $\hat{z}$

Result 10. A rotation about $\hat{z}$ cannot change S_z , so it leaves the S_z basis states where they are and can only multiply them by a phase:

$$R(\phi\hat{z})|\pm\hat{z}\rangle = e^{\mp i\phi/2}|\pm\hat{z}\rangle.$$

The exponent follows from $R = e^{-i\phi\hat{S}_z/\hbar}$ acting on a state with $S_z = \pm\hbar/2$.

Remark 26. A full turn, $\phi = 2\pi$, gives $R = -\mathbb{1}$ rather than $\mathbb{1}$: a spin- $\frac{1}{2}$ state needs 4π to come back to itself. The sign is a global phase, so it is unobservable on its own.

Remark 27. In the S_z basis the matrix is diagonal,

$$R(\phi\hat{z}) \xrightarrow{S_z} \begin{pmatrix} e^{-i\phi/2} & 0 \\ 0 & e^{i\phi/2} \end{pmatrix},$$

but in any other basis it is not, and it has to be built entry by entry from $R_{ij} = \langle i|R|j\rangle$.

Problem 8 ($R(\phi\hat{z})$ in the S_x basis). Find the matrix representation of $R(\phi\hat{z})$ in the S_x basis.

Remark 28. Repeating the calculation in the S_y basis, where the bras carry $\mp i$, gives the same matrix. Both $\hat{x}$ and $\hat{y}$ are perpendicular to the rotation axis, so the rotation looks identical from either one.

September 21,
2026

5.4.2 Rotation as a change of basis

Rotating the S_z basis by 90° about $\hat{y}$ turns it into the S_x basis, so we require

$$R\left(\frac{\pi}{2}\hat{y}\right)|\pm\hat{z}\rangle = |\pm\hat{x}\rangle.$$

Multiplying from the left by $\langle\pm\hat{z}|$ gives the matrix elements of $R\left(\frac{\pi}{2}\hat{y}\right)$ in the S_z basis:

$$R\left(\frac{\pi}{2}\hat{y}\right) \xrightarrow{S_z} \begin{pmatrix} \langle+\hat{z}|R|+\hat{z}\rangle & \langle+\hat{z}|R|-\hat{z}\rangle \\ \langle-\hat{z}|R|+\hat{z}\rangle & \langle-\hat{z}|R|-\hat{z}\rangle \end{pmatrix} = \begin{pmatrix} \langle+\hat{z}|+\hat{x}\rangle & \langle+\hat{z}|-\hat{x}\rangle \\ \langle-\hat{z}|+\hat{x}\rangle & \langle-\hat{z}|-\hat{x}\rangle \end{pmatrix}.$$

From $|\pm\hat{x}\rangle = \frac{1}{\sqrt{2}}(|+\hat{z}\rangle \pm |-\hat{z}\rangle)$ and orthonormality, $\langle+\hat{z}|\pm\hat{x}\rangle = \frac{1}{\sqrt{2}}$ and $\langle-\hat{z}|\pm\hat{x}\rangle = \pm\frac{1}{\sqrt{2}}$.

Result 11.

$$R\left(\frac{\pi}{2}\hat{y}\right) \xrightarrow{S_z} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

Remark 29. The columns of the matrix are the target kets written as S_z columns: the first column is $|\hat{x}\rangle$ and the second is $|\hat{y}\rangle$. The same shortcut gives any other change-of-basis rotation, for example the one taking $|\pm\hat{z}\rangle$ to $|\pm\hat{y}\rangle$ has columns $\frac{1}{\sqrt{2}}(1, \pm i)^T$.

Now take a state given in the S_x basis,

$$|\psi\rangle = \langle+\hat{x}|\psi\rangle|\hat{x}\rangle + \langle-\hat{x}|\psi\rangle|\hat{y}\rangle \xrightarrow{S_x} \psi^x = \begin{pmatrix} \langle+\hat{x}|\psi\rangle \\ \langle-\hat{x}|\psi\rangle \end{pmatrix}.$$

To find its S_z column, insert $\mathbf{1} = |\hat{x}\rangle\langle+\hat{x}| + |\hat{y}\rangle\langle-\hat{x}|$ into each S_z amplitude:

$$\langle\pm\hat{z}|\psi\rangle = \langle\pm\hat{z}|+\hat{x}\rangle\langle+\hat{x}|\psi\rangle + \langle\pm\hat{z}|\hat{y}\rangle\langle-\hat{x}|\psi\rangle.$$

In matrix form this reads

$$\psi^z = \begin{pmatrix} \langle+\hat{z}|\psi\rangle \\ \langle-\hat{z}|\psi\rangle \end{pmatrix} = \begin{pmatrix} \langle+\hat{z}|+\hat{x}\rangle & \langle+\hat{z}|\hat{y}\rangle \\ \langle-\hat{z}|+\hat{x}\rangle & \langle-\hat{z}|\hat{y}\rangle \end{pmatrix} \begin{pmatrix} \langle+\hat{x}|\psi\rangle \\ \langle-\hat{x}|\psi\rangle \end{pmatrix},$$

and the 2×2 matrix is exactly the one found above.

Result 12 (Change of basis).

$$\psi^z = R\left(\frac{\pi}{2}\hat{y}\right)\psi^x, \quad \psi^x = R^\dagger\left(\frac{\pi}{2}\hat{y}\right)\psi^z, \quad (43)$$

where the second follows from unitarity.

Result 13 (Expectation Value). With the state in the S_z basis, expectation values follow from

$$\langle S_z \rangle = \psi^{z\dagger} \hat{S}_z \psi^z. \quad (44)$$

Problem 9 (Expectation value from the S_x basis). For $|\psi\rangle = \frac{1}{\sqrt{3}}|\hat{x}\rangle + i\sqrt{\frac{2}{3}}|\hat{y}\rangle$, find $\langle S_z \rangle$ using matrix mechanics.

6 Change of Basis in n -Level Systems

6.1 Two bases for the same space

Take an n -level system with two orthonormal bases,

$$\begin{aligned} \{|j\rangle : j = 1, \dots, n\}, & \quad \{|\alpha\rangle : \alpha = 1, \dots, n\}, \\ \langle i|j\rangle = \delta_{ij}, & \quad \langle \alpha|\beta\rangle = \delta_{\alpha\beta}. \end{aligned}$$

Remark 30. The cross overlaps are not Kronecker deltas: $\langle j|\alpha\rangle \neq \delta_{j\alpha}$ in general, just as $\langle +\hat{x}|\hat{z}\rangle = \frac{1}{\sqrt{2}}$ is neither 0 nor 1.

Both bases are complete, so the identity can be inserted either way:

$$|\psi\rangle = \mathbb{1} |\psi\rangle = \sum_{j=1}^n |j\rangle \langle j|\psi\rangle = \sum_{\alpha=1}^n |\alpha\rangle \langle \alpha|\psi\rangle. \quad (45)$$

The same ket then has two column representations,

$$|\psi\rangle \xrightarrow{|j\rangle} \psi^j = \begin{pmatrix} \langle 1|\psi\rangle \\ \langle 2|\psi\rangle \\ \vdots \\ \langle n|\psi\rangle \end{pmatrix}, \quad |\psi\rangle \xrightarrow{|\alpha\rangle} \psi^\alpha = \begin{pmatrix} \langle \alpha=1|\psi\rangle \\ \langle \alpha=2|\psi\rangle \\ \vdots \\ \langle \alpha=n|\psi\rangle \end{pmatrix}. \quad (46)$$

6.2 The transformation matrix

Inserting $\mathbb{1} = \sum_{\alpha} |\alpha\rangle \langle \alpha|$ into each $|j\rangle$ amplitude,

$$\langle j|\psi\rangle = \sum_{\alpha=1}^n \langle j|\alpha\rangle \langle \alpha|\psi\rangle,$$

which is a matrix times a column.

Result 14 (Change of basis).

$$\psi^j = U \psi^\alpha, \quad U_{j\alpha} = \langle j|\alpha\rangle. \quad (47)$$

The columns of U are the new basis kets $|\alpha\rangle$ written in the $|j\rangle$ basis.

Theorem 2. U is unitary: $U^\dagger U = U U^\dagger = \mathbb{1}$.

Proof. Using $(U^\dagger)_{\alpha j} = U_{j\alpha}^* = \langle \alpha|j\rangle$ and completeness of $\{|j\rangle\}$,

$$(U^\dagger U)_{\alpha\beta} = \sum_j \langle \alpha|j\rangle \langle j|\beta\rangle = \langle \alpha|\beta\rangle = \delta_{\alpha\beta}.$$

The same argument with completeness of $\{|\alpha\rangle\}$ gives $U U^\dagger = \mathbb{1}$. $\square$

Remark 31. Normalization is basis-independent, as it must be:

$$\langle \psi|\psi\rangle = \psi^{j\dagger} \psi^j = \psi^{\alpha\dagger} U^\dagger U \psi^\alpha = \psi^{\alpha\dagger} \psi^\alpha = 1.$$

Example. With $|j\rangle = |\pm\hat{z}\rangle$ and $|\alpha\rangle = |\pm\hat{x}\rangle$, $U_{j\alpha} = \langle \pm\hat{z}|\pm\hat{x}\rangle$ is exactly $R(\frac{\pi}{2}\hat{y})$ from the previous section, and $\psi^j = U \psi^\alpha$ is $\psi^z = R \psi^x$.

6.3 Eigenstates and eigenvalues

Definition 38 (Eigenstate). A state $|a_i\rangle$ is an eigenstate of the operator $\hat{A}$ with eigenvalue a_i if

$$\hat{A}|a_i\rangle = a_i|a_i\rangle. \quad (48)$$

In matrix form this is the familiar $Mv = \lambda v$, with v the eigenvector.

Remark 32. A general operator maps a state to a different state, $\hat{A}|\psi\rangle = c|\phi\rangle$. Eigenstates are the special states that come back to themselves up to a number.

Example.

$$\hat{S}_z|\pm\hat{z}\rangle = \pm\frac{\hbar}{2}|\pm\hat{z}\rangle,$$

so $|\pm\hat{z}\rangle$ are the eigenstates of $\hat{S}_z$ and $\pm\hbar/2$ its eigenvalues, i.e. the measurement outcomes.

Proposition 15 (An operator is diagonal in its own eigenbasis).

$$A_{ij} = \langle a_i|\hat{A}|a_j\rangle = a_j\langle a_i|a_j\rangle = a_j\delta_{ij}, \quad (49)$$

so

$$\hat{A} \xrightarrow{|a_j\rangle} D = \begin{pmatrix} a_1 & 0 & \cdots & 0 \\ 0 & a_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_n \end{pmatrix}. \quad (50)$$

In particular $\langle a_i|\hat{A}|a_i\rangle = a_i$.

Example. $\hat{S}_z \xrightarrow{S_z} \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$.

6.4 Operators in another basis

Let $|j\rangle = |a_j\rangle$ be the eigenbasis of $\hat{A}$, so $\hat{A} \xrightarrow{|a_j\rangle} D$, and let $U_{j\alpha} = \langle a_j|\alpha\rangle$.

Proposition 16 (Similarity transformation).

$$\hat{A} \xrightarrow{|\alpha\rangle} A^\alpha = U^\dagger D U. \quad (51)$$

Proof. Insert the identity twice:

$$A_{\alpha\beta}^\alpha = \langle \alpha|\hat{A}|\beta\rangle = \sum_{i,j} \langle \alpha|a_i\rangle \langle a_i|\hat{A}|a_j\rangle \langle a_j|\beta\rangle = \sum_{i,j} (U^\dagger)_{\alpha i} D_{ij} U_{j\beta} = (U^\dagger D U)_{\alpha\beta}.$$

□

Result 15 (Expectation value in any basis).

$$\langle A \rangle = \langle \psi|\hat{A}|\psi\rangle = \psi^{j\dagger} D \psi^j = \psi^{\alpha\dagger} [U^\dagger D U] \psi^\alpha. \quad (52)$$

Either change the state to the eigenbasis ($\psi^j = U\psi^\alpha$) or change the operator to the state's basis ($A^\alpha = U^\dagger D U$). Both give the same number.

Problem 10 ($\hat{S}_z$ in the S_x basis). Using $U = R(\frac{\pi}{2}\hat{y})$, find the matrix of $\hat{S}_z$ in the S_x basis, and use it to redo the problem “Expectation value from the S_x basis” without changing the state’s basis.

A Appendix

B Solutions

Solution 1 (The exponential as a limit). Take the logarithm of the left side, which turns the power into a product:

$$\ln \left(1 + \frac{x}{N}\right)^N = N \ln \left(1 + \frac{x}{N}\right).$$

The Taylor series of the logarithm about $u = 0$ is

$$\ln(1 + u) = u - \frac{u^2}{2} + \frac{u^3}{3} - \cdots,$$

so with $u = x/N$, which is small once N is large,

$$N \ln \left(1 + \frac{x}{N}\right) = N \left(\frac{x}{N} - \frac{x^2}{2N^2} + \frac{x^3}{3N^3} - \cdots \right) = x - \frac{x^2}{2N} + \frac{x^3}{3N^2} - \cdots.$$

Every term after the first carries a factor $1/N$ or smaller, so they all vanish as $N \rightarrow \infty$ and the logarithm tends to x . Exponentiating,

$$\lim_{N \rightarrow \infty} \left(1 + \frac{x}{N}\right)^N = e^x.$$

Solution 2 (Other bases). The same state has different columns in different bases:

$$|+\hat{z}\rangle \xrightarrow{S_z} \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |+\hat{z}\rangle \xrightarrow{S_x} \begin{pmatrix} \langle +\hat{x} | +\hat{z} \rangle \\ \langle -\hat{x} | +\hat{z} \rangle \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad |+\hat{z}\rangle \xrightarrow{S_y} \begin{pmatrix} \langle +\hat{y} | +\hat{z} \rangle \\ \langle -\hat{y} | +\hat{z} \rangle \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}.$$

Solution 3 (A probability by matrix mechanics). Write $|\psi\rangle$ as a column and $\langle +\hat{x}|$ as the conjugate row:

$$\psi \xrightarrow{S_z} \frac{1}{\sqrt{5}} \begin{pmatrix} 1 \\ 2i \end{pmatrix}, \quad \langle +\hat{x} | \xrightarrow{S_z} \frac{1}{\sqrt{2}} (1 \quad 1).$$

Then

$$\langle +\hat{x} | \psi \rangle = \frac{1}{\sqrt{10}} (1 \quad 1) \begin{pmatrix} 1 \\ 2i \end{pmatrix} = \frac{1+2i}{\sqrt{10}}, \quad P(S_x = +\frac{\hbar}{2}) = \frac{|1+2i|^2}{10} = \frac{5}{10} = \frac{1}{2}.$$

Take the modulus of the whole sum, not of each term separately.

Solution 4 (The matrix of $\hat{S}_z$). Here every overlap is 0 or 1 by orthonormality, so only the diagonal survives:

$$\langle +\hat{z} | \hat{S}_z | +\hat{z} \rangle = \frac{\hbar}{2} (1 - 0) = \frac{\hbar}{2}, \quad \langle -\hat{z} | \hat{S}_z | -\hat{z} \rangle = \frac{\hbar}{2} (0 - 1) = -\frac{\hbar}{2},$$

with both off-diagonal entries zero, giving

$$\hat{S}_z \xrightarrow{S_z} \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

An operator is diagonal in its own basis, and the diagonal entries are its outcomes.

Solution 5 (The matrix of $\hat{S}_x$). From $|\pm\hat{x}\rangle \xrightarrow{S_z} \frac{1}{\sqrt{2}}(1, \pm 1)^T$ the overlaps are $\langle +\hat{z} | \pm\hat{x} \rangle = \langle -\hat{z} | \pm\hat{x} \rangle = \frac{1}{\sqrt{2}}$ and $\langle -\hat{z} | -\hat{x} \rangle = -\frac{1}{\sqrt{2}}$, all real. Then

$$\begin{aligned} \langle +\hat{z} | \hat{S}_x | +\hat{z} \rangle &= \frac{\hbar}{2} \left(\frac{1}{2} - \frac{1}{2} \right) = 0, & \langle +\hat{z} | \hat{S}_x | -\hat{z} \rangle &= \frac{\hbar}{2} \left(\frac{1}{2} + \frac{1}{2} \right) = \frac{\hbar}{2}, \\ \langle -\hat{z} | \hat{S}_x | +\hat{z} \rangle &= \frac{\hbar}{2} \left(\frac{1}{2} + \frac{1}{2} \right) = \frac{\hbar}{2}, & \langle -\hat{z} | \hat{S}_x | -\hat{z} \rangle &= \frac{\hbar}{2} \left(\frac{1}{2} - \frac{1}{2} \right) = 0, \end{aligned}$$

where the + signs come from the single minus sign carried by $\langle -\hat{z} | -\hat{x} \rangle$ meeting the minus in front of the second projector. So

$$\hat{S}_x \xrightarrow{S_z} \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Solution 6 (The matrix of $\hat{S}_y$). Now the overlaps are complex, and the bra side is conjugated: $\langle -\hat{z} | \pm\hat{y} \rangle = \pm \frac{i}{\sqrt{2}}$ while $\langle \pm\hat{y} | -\hat{z} \rangle = \mp \frac{i}{\sqrt{2}}$, with $\langle +\hat{z} | \pm\hat{y} \rangle = \langle \pm\hat{y} | +\hat{z} \rangle = \frac{1}{\sqrt{2}}$. Then

$$\begin{aligned} \langle +\hat{z} | \hat{S}_y | -\hat{z} \rangle &= \frac{\hbar}{2} \left(\frac{1}{\sqrt{2}} \cdot \frac{-i}{\sqrt{2}} - \frac{1}{\sqrt{2}} \cdot \frac{i}{\sqrt{2}} \right) = -\frac{i\hbar}{2}, \\ \langle -\hat{z} | \hat{S}_y | +\hat{z} \rangle &= \frac{\hbar}{2} \left(\frac{i}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} - \frac{-i}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} \right) = +\frac{i\hbar}{2}, \end{aligned}$$

and the diagonal entries vanish because $i \cdot (-i) = 1$ in both terms. So

$$\hat{S}_y \xrightarrow{S_z} \frac{\hbar}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}.$$

Solution 7 (Expectation value and uncertainty by matrix mechanics). First move to the S_z basis:

$$\psi \xrightarrow{S_z} \frac{1}{\sqrt{10}} \left[\begin{pmatrix} 1 \\ i \end{pmatrix} + \begin{pmatrix} 2i \\ 2 \end{pmatrix} \right] = \frac{1}{\sqrt{10}} \begin{pmatrix} 1 + 2i \\ 2 + i \end{pmatrix}.$$

Then $\langle S_z \rangle = \psi^\dagger \hat{S}_z \psi$:

$$\langle S_z \rangle = \frac{\hbar}{2} \cdot \frac{1}{10} \begin{pmatrix} 1 - 2i & 2 - i \end{pmatrix} \begin{pmatrix} 1 + 2i \\ -(2 + i) \end{pmatrix} = \frac{\hbar}{2} \cdot \frac{5 - 5}{10} = 0.$$

Since $\hat{S}_z^2 = \frac{\hbar^2}{4}\mathbb{1}$, we get $\langle S_z^2 \rangle = \frac{\hbar^2}{4}\psi^\dagger\psi = \frac{\hbar^2}{4}$, and therefore

$$(\Delta S_z)^2 = \frac{\hbar^2}{4} - 0, \quad \Delta S_z = \frac{\hbar}{2}.$$

Both S_z outcomes have probability $\frac{1}{2}$ here, so the spread is maximal.

Solution 8 ($R(\phi\hat{z})$ in the S_x basis). Apply R to the S_x basis states by expanding them in $\hat{z}$:

$$R|\pm\hat{x}\rangle = \frac{1}{\sqrt{2}} \left(e^{-i\phi/2} |+\hat{z}\rangle \pm e^{i\phi/2} |-\hat{z}\rangle \right).$$

Hitting these with $\langle\pm\hat{x}| = \frac{1}{\sqrt{2}} (\langle+\hat{z}| \pm \langle-\hat{z}|)$ gives

$$\begin{aligned} \langle+\hat{x}|R|+\hat{x}\rangle &= \langle-\hat{x}|R|-\hat{x}\rangle = \frac{1}{2} \left(e^{-i\phi/2} + e^{i\phi/2} \right) = \cos \frac{\phi}{2}, \\ \langle+\hat{x}|R|-\hat{x}\rangle &= \langle-\hat{x}|R|+\hat{x}\rangle = \frac{1}{2} \left(e^{-i\phi/2} - e^{i\phi/2} \right) = -i \sin \frac{\phi}{2}, \end{aligned}$$

using the two identities from the Taylor section. So

$$R(\phi\hat{z}) \xrightarrow{S_x} \begin{pmatrix} \cos \frac{\phi}{2} & -i \sin \frac{\phi}{2} \\ -i \sin \frac{\phi}{2} & \cos \frac{\phi}{2} \end{pmatrix}.$$

Check: at $\phi = \pi/2$, $R|+\hat{x}\rangle \xrightarrow{S_x} \frac{1}{\sqrt{2}}(1, -i)^T$, which is $|+\hat{y}\rangle$ up to a global phase, as rotating $\hat{x}$ by 90° about $\hat{z}$ should give.

Solution 9 (Expectation value from the S_x basis). The state in the S_x basis is

$$|\psi\rangle \xrightarrow{S_x} \psi^x = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ i\sqrt{2} \end{pmatrix}.$$

Change to the S_z basis:

$$\psi^z = R\left(\frac{\pi}{2}\hat{y}\right)\psi^x = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ i\sqrt{2} \end{pmatrix} = \frac{1}{\sqrt{6}} \begin{pmatrix} 1+i\sqrt{2} \\ 1-i\sqrt{2} \end{pmatrix}, \quad \psi^{z\dagger} = \frac{1}{\sqrt{6}} \begin{pmatrix} 1-i\sqrt{2} & 1+i\sqrt{2} \end{pmatrix}.$$

Then

$$\langle S_z \rangle = \psi^{z\dagger} \hat{S}_z \psi^z = \frac{\hbar}{2} \cdot \frac{1}{6} \begin{pmatrix} 1-i\sqrt{2} & 1+i\sqrt{2} \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1+i\sqrt{2} \\ 1-i\sqrt{2} \end{pmatrix} = \frac{\hbar}{2} \left(\frac{3}{6} - \frac{3}{6} \right) = 0.$$

Both S_z outcomes have probability $\frac{1}{2}$, since the two components have equal modulus.

C Useful Links

Stern-Gerlach experiment: <https://www.youtube.com/shorts/QG7kJXxN1Kk>